Dilution

Consider a 500 mL solution of salt water ($NaCl_{(aq)}$). If I dilute the solution by adding an additional 500 mL of pure water, have I changed the concentration? Have I changed the amount of salt present in the solution?

If the amount of salt does not change, then the moles remains constant.

$$\boxed{n_1 = n_2}$$

We know that $C = \frac{n}{V}$. Solving for *n*, we can show that *n* = *CV*

Therefore,

$$\boxed{C_1V_1 = C_2V_2}$$

This is our dilution formula

Ex) 2.00L NaOH solution is created from 750mL of 2.50 M NaOH stock solution. What is the concentration of the diluted solution?

$C_1V_1 = C_2V_2$

$(2.50)(0.75L) = C_2(2.00)$

$C_2 = 0.9375$

$= 0.94\ M$

$C_1V_1 = C_2V_2$

$(2.50)(750) = C_2(2000)$

$C_2 = \frac{(2.50\,mol/L)(750\,\cancel{mL})}{2000\,\cancel{mL}}$

Ex) How much 12.0 M $HCl_{(aq)}$ is required to make a 3.0 L solution of 1.0 M $HCl_{(aq)}$?

$C_1V_1 = C_2V_2$

$(12.0)V_1 = (1.0)(3.0)$

$V_1 = 0.25\ L$

Ex) 500.0 mL of a 6.5 M H_2SO_4 solution is diluted down to 1.5 M. How much water was added?

$C_1V_1 = C_2V_2$

$(6.5)(0.5) = (1.5)V_2$

$V_2 = 2.16$

$= 2.2\ L$

or

$2200\,mL$

$2200 - 500 = \underline{1700\,mL}$

Dilution Worksheet

p399 #1-3, 5, 8, 10, 25; p402 #7-12

p448 #17, 21, 22, 24-26; p450 #4-6, 17, 18